

KWORDINARYVQA: A KEYWORD-DRIVEN GENERATIVE VISUAL QUESTION ANSWERING SYSTEM FOR CULINARY EXPLORATION

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INTRODUCTION

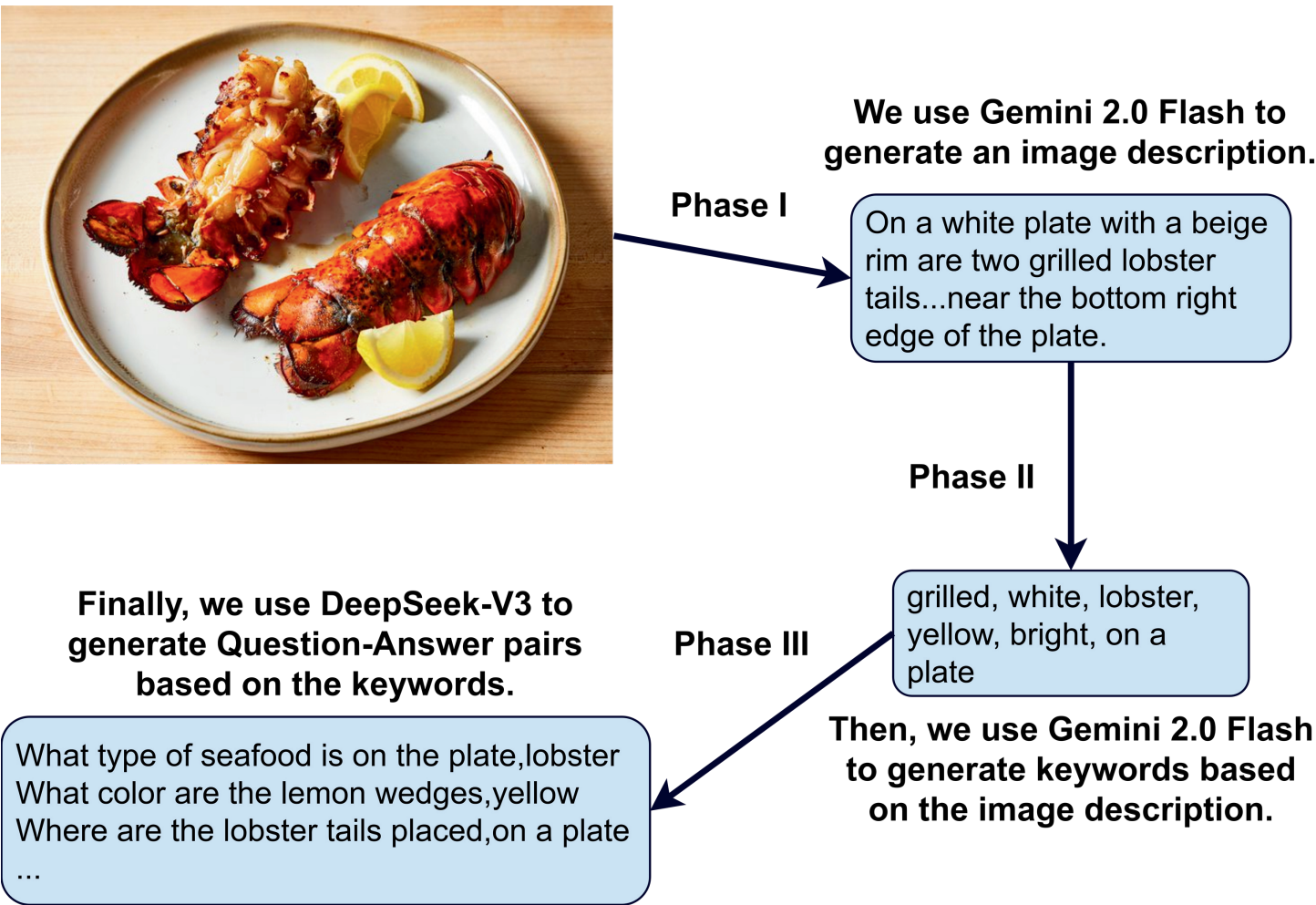
- General VQA models fail at the visual complexity and specific knowledge required for the food domain.
- There is a scarcity of large-scale, open-ended VQA datasets dedicated to food.
- We present KWordinaryVQA, a novel, automated pipeline to build a generative food VQA benchmark.

CONTRIBUTION

- Designed and implemented a fully automated pipeline for generating domain-specific VQA data.
- Introduced the **KWordinaryVQA dataset**, a challenging VQA benchmark in the food domain.
- Conducted a comprehensive benchmark of diverse VQA methods, including retrieval baselines, fine-tuned classifiers, and generative LLMs.

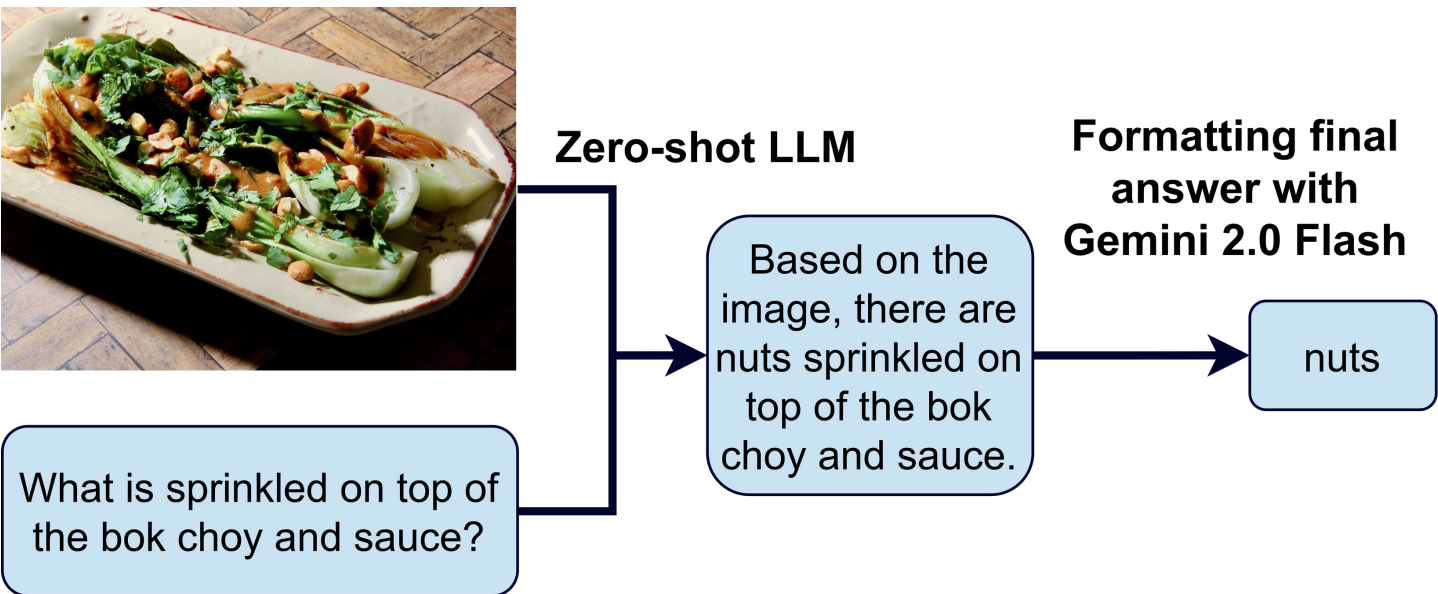
METHODOLOGY

The 3-stage automated dataset generation.



- Our methodology began by curating 8,693 images from 49,332 raw Allrecipes entries. This involved a multi-step preprocessing pipeline:
 - Fuzzy string matching for name normalization.
 - A two-tiered undersampling strategy to balance classes (e.g., common dishes, vegan/non-vegan).
 - Scope-based filtering for linguistic consistency.
- These curated images were then processed by our automated pipeline (shown in Figure 1).

The two-phase post-processing pipeline.



DATA ANALYSIS

Table 1: Dataset statistic.

Statistic	Count
# Entries	42,750
# Unique questions	29,822
# Unique answers	4,413
# Images	8,693

Figure 1: Question types.

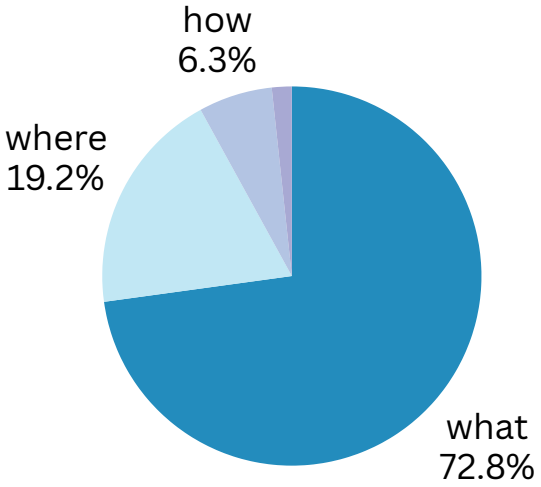
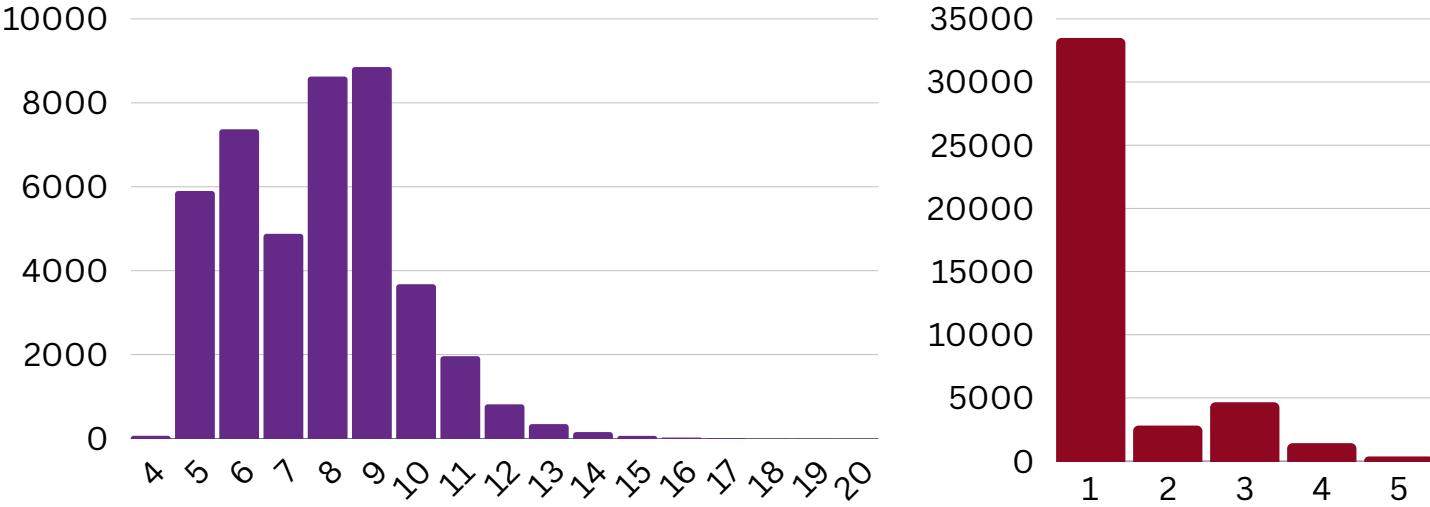


Figure 2: Distribution of question and answer lengths.



EXPERIMENTS

Table 2: Key benchmark results.

Model	Accuracy	F1-score	GPTScores
BEiT-3 (fine-tuned)	0.4804	0.5455	0.6864
Gemini 2.0 Flash (zero-shot with post-processed)	0.3533	0.5130	0.7239
MiniCPM-o 2.6 (few-show)	0.3225	0.4761	0.6702

Our benchmark confirms the dataset's challenge: top models' low accuracy and high GPTScores mask a deeper gap in the domain-specific culinary knowledge required for precise answers.

CONCLUSION

We introduced a novel, end-to-end automated pipeline for generating domain-specific VQA data, which we used to create KWordinaryVQA, a challenging new benchmark for the food domain. Our benchmark reveals a critical accuracy-semantic trade-off, underscoring the dataset's difficulty.

REFERENCES

Fu, J., Ng, S.K., Jiang, Z., and Liu, P. 2024. GPTScores: Evaluate as you desire. In NAACL 2024.
Google DeepMind. 2025. Gemini 2.0 flash. Accessed: 2025-03-25.
Goyal, Y., Khot et al. 2017. Making the V in VQA matter: Elevating the role of image understanding in Visual Question Answering. In CVPR 2017
Liu, A., et al. 2024. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*.
Wang, W., et al. 2023. Image as a foreign language: Beit pretraining for vision and vision-language tasks. In CVPR 2023.

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